Event-Driven Gain in Expectancy (EDGE): A Probabilistic Metric for Comprehensive Player Performance in Tennis

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Abstract

This work proposes a new tennis metric that summarizes a player's overall performance in a single measure. The proposed metric, called Event-Driven Gain in Expectancy (EDGE), is based on a simple concept: not all point-level events are equally valuable. EDGE estimates the value of each event type, such as aces, winners, double faults, and unforced errors, and uses these values to weight the positive and negative events produced by a player. By combining multiple point-level events into a single numerical score, EDGE enables direct comparison of overall performance across players. Experimental results show that EDGE captures point-level performance that is related to, but not fully represented by, the tour ranking.

X.1 Introduction

Tennis produces a wide range of statistics to describe player performance. Common measures used in TV broadcasts and tour websites include aces, double faults, winners, unforced errors, first-serve percentage, and break-point conversion rate. While these measures are intuitive and useful because each captures a specific aspect of play, they are narrowly focused. For example, an ace measures one aspect of serving, a winner captures one type of point-winning outcome, and an unforced error represents one type of unsuccessful outcome. As a result, for assessing a player's overall performance, it is often required to examine many statistics simultaneously, typically through tables, radar charts, or other visualizations. This makes interpretation and comparison less direct.

Composite metrics address this limitation by combining multiple aspects of play into a single numerical measure. For instance, Dominance Ratio (DR) and Hold-plus-break percentage (H+B) summarize performance by combining serve and return outcomes. However, these metrics are deterministic summaries of observed outcomes and do not explicitly incorporate score context

into their calculations. Therefore, they do not distinguish between, for example, the score-dependent impact of a winner on break point and that of a winner at 40–0, even though the two outcomes can have different effects on the probability of winning the game.

This project introduces Event-Driven Gain in Expectancy (EDGE), a probabilistically derived composite metric for summarizing player performance. Using tour-wide point-by-point match data, EDGE estimates weights for positive and negative point-level event types, such as aces, winners, double faults, and unforced errors, based on their average impact on the probability of winning the current game. To derive these weights, EDGE first evaluates individual event occurrences using score-dependent changes in game-win probability and then averages those changes by event type. A player's EDGE is then calculated as a linear weighted sum of event counts, normalized by the total number of points played. Higher EDGE indicates that a player produces a more valuable mix of positive and negative events, based on their tour-wide average effects on the probability of winning the current game. In this way, EDGE combines familiar tennis statistics probabilistically rather than treating each event type as an isolated count.

This chapter presents the motivation, design, and initial evaluation of EDGE. It first reviews existing performance metrics, with particular attention to their treatment of score context. It then describes the probabilistic formulation of EDGE, including the estimation of game-win probability and the derivation of event weights from point-by-point match data. Experimental results examine the empirical characteristics of EDGE and its relationship with tour rankings with professional women’s match data.

X.2 Background

This section overviews the tennis scoring system, describes the concept of game-win expectancy, and reviews existing performance metrics.

X.2.1 Scoring System in Tennis

A tennis match consists of sets, games, and points. A set consists of a sequence of games. Two players alternate service between games, with one player serving throughout each game. A set is won by the first player to win at least six games with a margin of two games. If the game count reaches 6-6, a tiebreak determines the winner of the set. In women’s tennis, a match is typically played as a best-of-three-set match, which means the first player to win two sets wins the match.

Each game consists of a sequence of points. Unlike ordinary numerical scoring, the score within a game progresses from 0, 15, 30, and 40 as a player wins points. A player wins a game by winning at least four points with a margin of two points. If both players reach 40, the score is called deuce. A player who wins the next point gains advantage; winning the following point

wins the game, while losing it returns the score to deuce. When the returning player is one point away from winning the game, the point is called a break point because winning it would "break" the opponent's serve.

X.2.2 Theoretical Game-win Expectancy

It is well known that the probability of winning a game changes substantially across score states [1–5]. Assuming the server wins each point independently with a constant probability p, a tennis game can be modeled as a Markov chain, which can derive the probability of eventually winning the game from any score state [1]. For example, Gutiérrez-Santiago et al. report empirical p values by analyzing points from the quarterfinals and later rounds of three 2021 Grand Slams in professional women's singles: p=0.61 on hard court (U.S. Open), p=0.62 on grass court (Wimbledon), and p=0.56 on clay court (Roland Garros) [5]. Klaassen and Magnus report similar results from Gland Slam matches in 2010: p=0.58 on hard court (U.S. Open), p=0.59 on grass court (Wimbledon), and p=0.56 on clay court (Roland Garros) [4].

With p=0.61, Fig. 1 shows the theoretical probability that the server wins the current game, hereafter referred to as *game-win expectancy* (GE), from each score state. At the start of a game (0–0), the server has 75.62% GE. In other words, the returner has a 24.38 % probability (100 – 75.62) of eventually winning the game. As the game progresses, GE changes with the score state.

		Returner score			
		0	15	30	40
Server score	0	75.62	59.95	38.98	16.11
	15	85.64	73.36	53.60	26.41
	30	93.49	85.99	70.98	43.30
	40	98.28	95.59	88.68	70.98

Fig. 1: Theoretical Game-win Expectancy (GE, %) for Score States, p=0.61

When the server leads 40–0, GE is very high, 98.28%, because only one additional point is needed to close the game, while the returner must win three consecutive points to reach deuce. At deuce, by contrast, the game is much less settled: either player must win two consecutive points to take the game, so the GE is closer between the server and returner (70.98%). As a result, the same point-level event can produce a different change in GE depending on the score state in which it occurs. For instance, a winner at 40–0 and a winner at deuce are both recorded simply as winners in conventional statistics, but they produce as much as 27.3% difference (98.28 – 70.98) in GE because they occur in different score situations.

X.2.3 Existing Composite Statistics and Tour Ranking

Dominance Ratio (DR) is a composite metric that summarizes a player's point-level dominance by combining serve and return performance [6, 7]. It is calculated as the percentage of return points won divided by the percentage of serve points lost. As a player wins more serve points, DR's denominator decreases, which increases DR. As a player wins more return points, DR's numerator increases, which also increases DR. Therefore, a higher DR means stronger overall performance across serve and return at individual points. A DR greater than 1 indicates that a player wins return points more often than losing serve points, suggesting overall point-level dominance over opponents. Conversely, a DR below 1 indicates overall point-level disadvantage against opponents. A DR of 1 indicates a neutral point-level balance.

Hold-plus-break percentage (H+B) is a composite metric that summarizes a player's game-level performance by combining serve and return results [8]. It is calculated by adding the percentage of service games won, or *hold percentage*, to the percentage of return games won, or *break percentage* [9]. As a player wins more service games, hold percentage increases, which increases H+B. As a player wins more return games, break percentage increases, which also increases H+B. Therefore, a higher H+B indicates stronger combined serve-and-return performance at the game level. An H+B greater than 100% indicates that a player wins service games at a higher rate than they lose return games, suggesting overall game-level dominance over opponents. Conversely, an H+B below 100% indicates an overall game-level disadvantage. An H+B of 100% represents a neutral balance between holding serve and losing return games.

DR and H+B are deterministic metrics because they are computed directly from observed outcomes. They do not account for the context in which those outcomes occur. In particular, DR treats points equally regardless of score state, so a point won at 40–0 contributes in the same way as a point won on break point. Similarly, H+B treats each service or return game as simply won or lost without considering how the underlying points were played or which score states they occurred at. In contrast, EDGE is a probabilistically derived composite metric that incorporates score context in estimating tour-average values for point-level events and weights a player's events by their estimated values. Therefore, EDGE can represent player performance in greater detail by differentiating point-level events according to their probabilistic impact.

A professional tour ranking orders players by the ranking points they have earned in eligible tournaments over a rolling 52-week period. Separate rankings are maintained for men by the Association of Tennis Professionals (ATP) and for women by the Women's Tennis Association (WTA). Players earn ranking points based on how far they advance in a tournament and the tournament's category. For example, in the WTA ranking system, winning one of the four Grand Slam tournaments (U.S. Open, Wimbledon, Roland Garros, or Australian Open) earns 2,000 points. Each of 10 WTA 1000 tournaments, including the Indian Wells Open, Miami Open, and

Cincinnati Open, awards 1,000 points to the winner. The 2026 WTA calendar also includes 17 WTA 500 tournaments and 22 WTA 250 tournaments, where the winners earn 500 and 250 points, respectively.

A tour ranking represents competitive success, but because ranking points are awarded according to tournament advancement and category, it does not directly measure a player's performance at the point, game, or match level. For example, two players who reach the same round of a tournament receive the same number of ranking points even if one advances through dominant straight-set victories while the other wins several close matches. Rankings can also be influenced by the number and categories of tournaments entered, the strength of opponents and draws, and injuries. As a result, a tour ranking may not fully reflect differences in players' on-court performance. In contrast, EDGE is based on what happens within matches rather than on tournament advancement or category. It is designed to characterize on-court performance by considering the probabilistic value of how points are won and lost.

X.3 EDGE: Event-Driven Gain in Expectancy

This section describes the algorithmic design of EDGE. EDGE measures a player's on-court performance by weighting point-level events according to their tour-wide average effect on the probability of winning the current game. Since games are the fundamental building blocks of a match, EDGE uses game-win expectancy (GE)—the probability of winning the current game from a given score state—as the basis for event valuation, rather than set- or match-win expectancy.

EDGE considers eight types of point-level events, as shown in Fig. 2. They span the serve, return, and rally phases of a point and include both positive and negative outcomes.

Phase	Positive event type	Negative event type
Serve	Ace Forced return error drawn	Double fault
Return	Return winner	Unforced return error
Rally	Rally winner Forced error drawn	Unforced error

Fig. 2: Event Types considered in EDGE

A forced return error drawn is credited to the server whose serve causes the return error. A forced error drawn is credited to the player whose shot causes the opponent's forced error.

X.3.1 Player-level EDGE Calculation

For a given player *p*, EDGE is computed as a linear weighted sum of event counts, normalized by the total number of points played:

$$\mathrm{EDGE}(p) = \frac{1}{P}\sum_{i=1}^{N} w_i c_i$$

where *N* is the total number of event types (*N*=8), c_i is the count of event type *i* for player *p*, w_i is the tour-wide weight assigned to event type *i*, and *P* is the total number of points played by player *p*, including points that do not end with one of the eight event types in Fig. 2. Thus, c_i and *P* vary by player, whereas the same tour-wide event weights w_i are applied to all players.

To calculate c_i and *P* and estimate w_i, this work uses crowdsourced point-by-point data from the Match Charting Project (MCP), hosted by Tennis Abstract [10]. Each w_i is derived in two steps: (1) empirically estimating game-win expectancy (GE) for each possible score state and (2) calculating the change in GE associated with each observed event and averaging these changes by event type.

X.3.2 Estimation of Game-Win Expectancy

In the first step, EDGE considers all 18 non-terminal score states, including 0-0, 15-0, 30-0, 40-0, 40-40 (deuce), advantage server (A-40), and advantage returner (40-A), and empirically estimates GE for each score state. For a score state *s*, the server's GE is defined as:

$$\mathrm{GE}(s) = P(\text{server wins game} \mid \mathrm{s}) = \frac{N_{\mathrm{win}}(s)}{N(s)}$$

where $N_{win}(s)$ counts occurrences of state *s* that eventually result in a server game win, and $N(s)$ counts all occurrences of state *s*. The numerator and denominator are calculated from the MCP data. Repeated visits to the same score state within a game, such as multiple returns to deuce, are counted as separate occurrences. By definition, GE is expressed from the server's perspective: $GE(s)$ is the probability that the server eventually wins the current game from score state *s*. The probability that the returner wins the game is $1 - GE(s)$.

X.3.3 Derivation of Event Weights

In the second step, EDGE uses these GE estimates to derive a weight for each event type. For each observed point in the MCP data, the change in GE is calculated from the perspective of the player associated with the point-ending event. Since GE is defined from the server's perspective, $GE(s)$ is used when that player is serving and $1 - GE(s)$ is used when that player is returning. The GE change is then calculated as the difference between the player-specific GE after and before the point. Note that the same event types occurring at different score states can produce different GE changes because GE varies across score states.

EDGE averages these GE changes across all occurrences of each event type to obtain its tour-wide event weight w_i:

$$w_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \Delta \mathrm{GE}_{ij}$$

where n_i is the tour-wide count of event type *i* and ΔGE_{ij} denotes the player-perspective change in GE for the *j*-th occurrence of event type *i*. Thus, w_i represents the average GE change associated with event type *i* across the tour. Positive event types shown in Fig. 2 are expected to have positive weights, while negative event types are expected to have negative weights; the magnitude of w_i reflects the average impact of the event type on GE.

Note that score context affects the estimation of w_i, but not the subsequent player-level aggregation (Eq. 1). Once a tour-wide weight has been estimated for an event type, every occurrence of that event type receives the same weight in the final EDGE calculation. Thus, an ace at 40–0 and an ace at deuce can produce different GE changes during weight estimation, but both use the same final ace weight when calculating a player's EDGE.

This separation is intentional. By averaging score-dependent GE changes into tour-wide weights before player-level aggregation, EDGE is designed as a predictive measure of future performance rather than a contextual or situational measure of past performance. It uses fixed event weights to reduce its dependence on the particular score situations in which past events occurred and instead emphasizes the types and frequencies of events the player produces.

X.4 Experimental Results

This section first estimates tour-wide game-win expectancy (GE) and event weights using professional women's match data and then computes EDGE for the top 100 players in the Women's Tennis Association (WTA) rankings as of May 25, 2026. It examines the empirical characteristics of GE, event weights, and EDGE, as well as the relationship between EDGE and

the WTA ranking. All experiments use the MCP point-by-point dataset for matches played in the 2020s [10]. This dataset contains 326,440 points from 2,399 matches played between January 6, 2020, and May 24, 2026. The Python programs used in the experiments are available in [11].

X.4.1 Empirical Game-win Expectancy (GE)

Fig. 3 shows the empirical GE for each score state, estimated from the MCP data. The "total count" column indicates how many times each score state occurred, while the "game-win count" column indicates how many times the server ultimately won the game from that state. GE is calculated as the percentage of server game wins. Fig. 4 presents the same results sorted in descending order of GE.

Figs. 3 and 4 show that GE varies substantially across score states, ranging from 10.10% when the server trails 0–40 to 97.30% when the server leads 40–0. Across the 18 score states, the mean and median GE are 60.22% and 63.68%, respectively. At tied score states, GE is 66.37% at 0–0, 64.75% at 15–15, 62.61% at 30–30, and 62.35% at deuce. These results indicate the general advantage of serving.

GE generally increases as the server advances toward winning the game and decreases as the returner advances. Importantly, the exact score state affects the magnitude of these changes. For example, winning a point at 40–0 increases the server's GE from 97.30% to 100%, a gain of only 2.70 percentage points. In contrast, winning a point at deuce increases GE from 62.35% to 83.62%, a gain of 21.27 percentage points. This result empirically justifies the motivation for incorporating score context when estimating EDGE event weights: identical point-level events can have very different effects on GE depending on when they occur.

The empirical GE estimates also show a pattern consistent with the theoretical estimates in Fig. 1. The relative ordering of score states is nearly identical. However, the empirical estimates are consistently lower than the theoretical estimates obtained with p=0.61. For example, GE at 0–0 is 66.37% empirically compared with 75.62% theoretically, while GE at deuce is 62.35% empirically compared with 70.98% theoretically. The gap may result from differences in the assumptions and data used to obtain the theoretical and empirical estimates. The theoretical model assumes a fixed probability of winning each service point, estimated with Grand Slam matches from the quarterfinals onward on hard court, whereas the empirical estimates aggregate observed outcomes across players of different strengths, matches, and surfaces.

Score state	Total count	Game-win count	Game-win expectancy
0-0	50,865	33,760	66.37
0-15	21,558	10,523	48.81
0-30	9,485	2,699	28.46
0-40	4,128	417	10.10
15-0	28,714	22,900	79.75
15-15	24,184	15,660	64.75
15-30	15,570	6,693	42.99
15-40	8,984	1,728	19.23
30-0	16,601	15,063	90.74
30-15	20,675	16,683	80.69
30-30	17,511	10,964	62.61
30-40	12,529	4,331	34.57
40-0	9,893	9,626	97.30
40-15	15,742	14,717	93.49
40-30	16,285	13,650	83.82
40-40	26,858	16,747	62.35
AD-40	15,268	12,767	83.62
40-AD	11,589	3,979	34.33

Fig. 3: Game-win Expectancy (GE, %) for Score States, Sorted by Score State

Score state	Total count	Game-win count	Game-win expectancy
40-0	9893	9626	97.30
40-15	15742	14717	93.49
30-0	16601	15063	90.74
40-30	16285	13650	83.82
AD-40	15268	12767	83.62
30-15	20675	16683	80.69
15-0	28714	22900	79.75
0-0	50865	33760	66.37
15-15	24184	15660	64.75
30-30	17511	10964	62.61
40-40	26858	16747	62.35
0-15	21558	10523	48.81
15-30	15570	6693	42.99
30-40	12529	4331	34.57
40-AD	11589	3979	34.33
0-30	9485	2699	28.46
15-40	8984	1728	19.23
0-40	4128	417	10.10

Fig. 4: Game-win Expectancy (GE, %) for Score States, Sorted by GE

X.4.2 Empirical Event Weights

Fig. 5 shows the tour-wide weights for the eight event types listed in Fig. 2, estimated from the MCP data. Positive weights indicate event types that, on average, increase a player's GE, and negative weights indicate those that decrease it. For example, a weight of 0.191 corresponds to an average increase of 19.1 percentage points in the player's GE. The "count" column indicates how many times each event type occurred.

Event type	Weight	Count
Return winner	0.191	7,607
Forced error drawn	0.180	57,690
Rally winner	0.173	72,630
Forced return error drawn	0.148	46,233
Ace	0.143	15,060
Unforced return error	– 0.149	12,583
Unforced error	– 0.185	97,310
Double fault	– 0.195	15,974

Fig. 5: Tour-wide Event Weights

Both positive and negative event weights fall within a relatively compact range. Positive events increase GE by about 14–19 percentage points, and negative events decrease GE by about 15–20 percentage points. Among negative events, double faults have the largest negative impact (–0.195), followed by unforced errors (–0.185). This is intuitive because both immediately surrender the point. Among positive events, return winners have the largest weight (0.191), followed by forced errors drawn (0.180) and rally winners (0.173), while aces have the smallest positive weight (0.143).

Note that, because these weights are averages of score-dependent GE changes, their differences reflect not only the event type itself but also the score states in which each event type tends to occur. Therefore, for example, a return winner is not intrinsically more "important" or "valuable" than an ace simply because it has a larger weight.

An intuitive explanation for the relatively low ace weight is that aces can occur often in lower-pressure score states such as 40–0, 40–15, and 30–0, in part because players tend to attempt bigger serves in such situations. Conversely, aces may occur less often in higher-pressure points, such as deuce and break points. Empirical evidence supports this pattern: Meffert et al. found that players served fewer aces on break points (8.53%) than on regular points (9.59%) at Wimbledon [12]. The MCP data show a similar pattern. As shown in Fig. 6, ace percentages are highest at lower-pressure score states, including 40–0 (8.13%), 40–15 (7.19%), and 30–0

(6.95%), while they are lower at break-point states, ranging from 3.22% at 0–40 to 4.21% at 40-A.

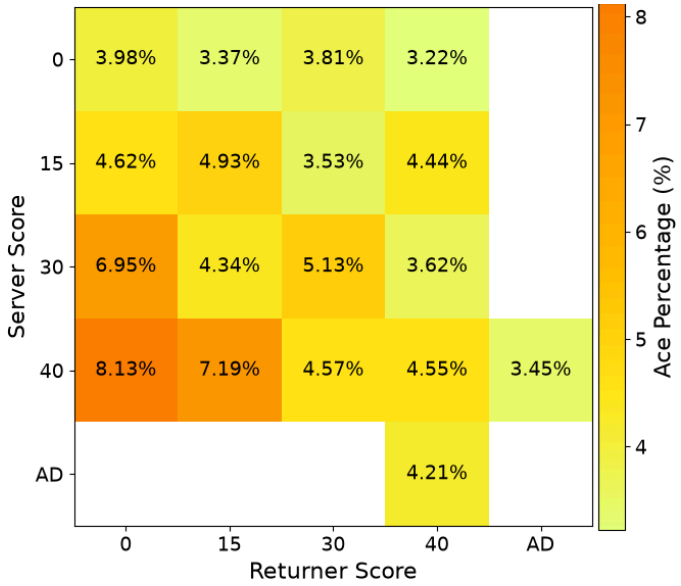


Fig. 6: Empirical Ace Percentage by Score State

The effect of this pattern becomes clear from the GE estimates in Fig. 3. When an ace occurs at 40–0, 40–15, or 30–0, the server's GE increases by only 2.70 (100–97.30), 6.51 (100–93.49), and 6.56 (97.30–90.74) percentage points, respectively. In contrast, an ace at deuce increases GE by 21.27 percentage points (83.62–62.35). Thus, an ace can be extremely valuable in a high-leverage situation even though the average ace weight is relatively low. Since the ace weight averages these score-dependent changes across all observed aces, ace occurrences in lower-pressure, lower-leverage situations contribute smaller GE gains and reduce the overall average weight.

X.4.3 EDGE of WTA Top 100 Players

Fig. 7 shows the distribution of EDGE values among WTA top-100 players, as of May 25, 2026, who had at least 15 matches recorded in the MCP data (n=59). For visualization, EDGE values are multiplied by 1,000.

The scaled EDGE values are distributed around a mean of 17.36 and a median of 17.62. The similarity between these values indicates little overall skewness. The middle 50% of players fall between 13.79 and 21.90, while the distribution extends beyond this central range in both directions. The lowest and highest scaled EDGE values are 2.34 and 32.15, respectively. No players fall outside the IQR-based outlier bounds of 1.62 and 34.08. Overall, the distribution is approximately unimodal, with most players concentrated in the middle range and fewer players

toward the extremes. This shows that variation in EDGE is spread across players rather than being driven by a few extreme values.

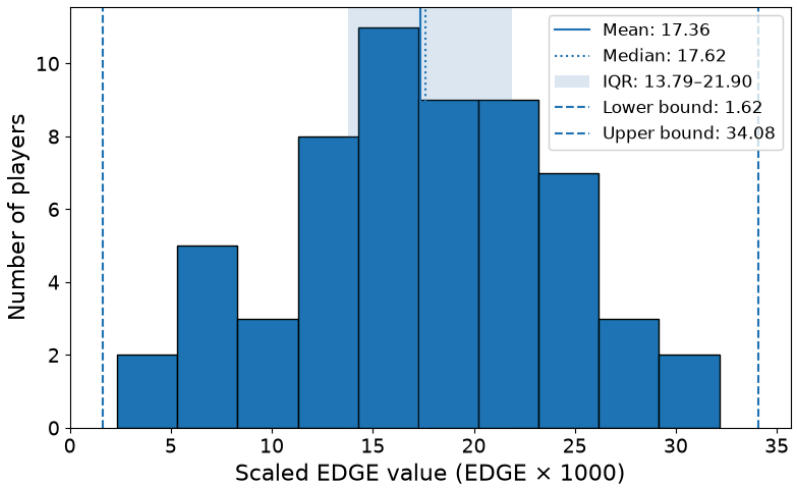


Fig. 7: Distribution of Scaled EDGE Values for WTA Top-100 Players (≥ 15 MCP Matches, *n*=59)

	EDGE (×1000)		WTA Ranking
1	Aryna Sabalenka	32.10	Aryna Sabalenka
2	Belinda Bencic	30.80	Elena Rybakina
3	Elena Rybakina	28.93	Iga Swiatek
4	Talia Gibson	27.61	Coco Gauff
5	Sorana Cirstea	26.28	Jessica Pegula
6	Alexandra Eala	24.86	Amanda Anisimova
7	Leylah Fernandez	24.31	Elina Svitolina
8	Iga Swiatek	24.14	Mirra Andreeva
9	Qinwen Zheng	24.01	Victoria Mboko
10	Sofia Kenin	23.84	Karolina Muchova
11	Jelena Ostapenko	23.70	Belinda Bencic
12	Magda Linette	23.27	Linda Noskova
13	Jessica Pegula	23.19	Jasmine Paolini
14	Liudmila Samsonova	22.45	Ekaterina Alexandrova
15	Linda Noskova	22.03	Marta Kostyuk
16	Solana Sierra	21.78	Naomi Osaka
17	Karolina Muchova	21.72	Iva Jovic
18	Anna Kalinskaya	21.30	Sorana Cirstea
19	Amanda Anisimova	20.86	Madison Keys
20	Clara Tauson	20.39	Clara Tauson

Fig. 8: Top-20 Overlap between EDGE and WTA Rankings

X.4.4 Relationship with the WTA Ranking

Fig. 8 compares the EDGE ranking with the WTA ranking for 59 eligible players in the WTA top 100 who had at least 15 matches recorded in the MCP data. Of the EDGE top-20 players, 10 are also in the WTA top 20. They are shaded in Fig. 8.

Among the 59 eligible players, 20 are in the WTA top 20. If 20 players were selected randomly from the 59 eligible players, the expected number of WTA top-20 players in this group would be 20×(20/59)=6.78 players. Thus, the observed overlap is about 47% larger than the random expectation (10/6.78). This overlap indicates that EDGE and the WTA ranking agree on a considerable number of high-performing players, despite being constructed from different types of performance data. At the same time, only half of the EDGE top 20 are in the WTA top 20, suggesting that EDGE does not simply reproduce the WTA ranking.

This interpretation is further supported by the rank correlation analysis (Fig. 9). The Spearman rank correlation between EDGE rank and WTA rank is rho=0.395 with p=0.0019, which indicates a moderate positive association between the two rankings. Overall, Figs. 8 and 9 show that EDGE and the WTA ranking are related but are not equivalent measures of player performance. EDGE evaluates how points are won and lost through positive and negative events, whereas the WTA ranking is based on ranking points earned through tournament advancement.

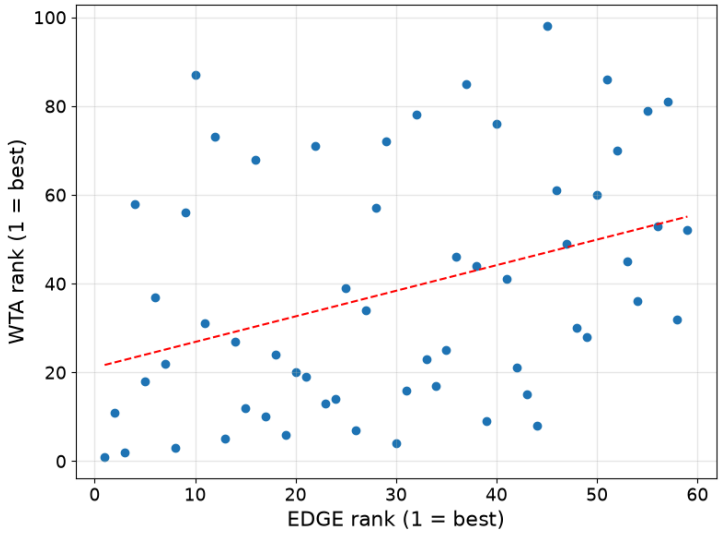


Fig. 9: Correlation between EDGE Rank and WTA Rank

Individual players also illustrate both the similarities and differences between the rankings. Aryna Sabalenka is ranked #1 by both measures, while Belinda Bencic (EDGE #2, WTA #11), Elena Rybakina (EDGE #3, WTA #2), Iga Swiatek (EDGE #8, WTA #3), and Jessica Pegula (EDGE #13, WTA #5) also rank highly under both measures (Fig. 8).

In contrast, several players outside the WTA top 20 appear in the EDGE top 20, including Talia Gibson (EDGE #4, WTA #58), Alexandra Eala (EDGE #6, WTA #37), Qinwen Zheng (EDGE #9, WTA #56), Sofia Kenin (EDGE #10, WTA #87), Magda Linette (EDGE #12, WTA #73), and Solana Sierra (EDGE #16, WTA #68).

Conversely, several WTA top-20 players fall outside the EDGE top 20, including Coco Gauff (WTA #4, EDGE #30), Mirra Andreeva (WTA #8, EDGE #44), Victoria Mboko (WTA #9, EDGE #39), and Marta Kostyuk (WTA #15, EDGE #43). Some of these discrepancies involve players with relatively few matches in the MCP data and therefore be more sensitive to sampling variation. Across the 59 eligible players, the number of MCP matches ranges from 15 to 218. For example, Sierra and Gibson have 15 and 16 matches, respectively.

Bencic ranks #2 by EDGE, compared with #11 in the WTA ranking. This discrepancy may result from her career interruption and the different time windows used by the two measures. In this work, EDGE is calculated with MCP data from January 2020 through May 2026, while the WTA ranking is based on ranking points earned during a rolling 52-week period. Bencic reached a career-high WTA ranking of #4 in February 2020 before taking maternity leave in September 2023. She returned to competition in October 2024, began 2025 ranked at #421, and climbed back to #11 by May 2026. Thus, her WTA ranking primarily represents her recent post-maternity results, whereas her EDGE ranking combines both pre- and post-maternity performance. Her #2 EDGE ranking therefore represents both her high level of play before the interruption and her strong performance after returning to competition. Bencic illustrates how EDGE can highlight strong point-level performance that may not be fully reflected in a player's WTA ranking.

Bencic's point-level performance is characterized by an exceptionally low unforced-error rate and strong contributions from positive events that force opponents into errors. Alexandra Eala (EDGE #6, WTA #37) shows a similar profile. Among the 59 eligible players, Bencic and Eala have the lowest and second-lowest unforced-error rates, at 11.0% and 11.5% of points, respectively, compared with an average of 14.7%. Since unforced errors have a relatively large negative weight (Fig. 5), reducing their frequency has a strong positive effect on a player's overall EDGE. On the positive side, Bencic has the highest forced-return-error-drawn contribution to EDGE among the 59 players and ranks #9 in return-winner contribution. Eala has the second-highest forced-error-drawn contribution. Eala's lower overall EDGE is partly due to weaker contributions from several other positive event types, including aces. As of September 14, 2026, Bencic and Eala were ranked #12 and #18 in the WTA rankings, respectively. Eala's WTA rank improved by 19 places, from #37 to #18, between May and September 2026.

Jessica Pegula (EDGE #13, WTA #5) and Sorana Cirstea (EDGE #5, WTA #18) also exhibit small negative-event contributions to EDGE, but they achieve this through different event profiles. Pegula ranks #3 among the 59 eligible players in minimizing negative-event

contributions, behind Eala and Bencic, driven by a low unforced-error rate (#3) and a relatively low double-fault rate (#9). Cirstea ranks #6 for negative-event performance, but unlike Bencic and Eala, her advantage comes less from unforced errors and more from avoiding double faults and unforced return errors, for which she ranks #2 and #3, respectively. On the positive side, Cirstea has a relatively balanced profile; her positive EDGE (#19) is lower than Bencic's (#9) but higher than Eala's (#29). Pegula's positive EDGE is lower (#35), with her rally-winner contribution particularly lower than those of Bencic, Eala, and Cirstea. Thus, Pegula's lower overall EDGE results from a weaker positive-event profile despite excellent control of negative events. As of September 14, 2026, Pegula and Cirstea were ranked #3 and #15 in the WTA rankings, respectively, up two and three places from their May rankings.

X.5 Future Work

Future work can extend EDGE in both its algorithmic formulation and its empirical evaluation. One algorithmic extension is to develop surface-specific EDGE measures. The current method estimates game-win expectancy (GE) and event weights using matches recorded across multiple surfaces. However, the frequencies and effects of point-level events can differ among hard, clay, and grass courts. Separate GE estimates and event weights for each surface could therefore produce surface-specific EDGE scores and characterize how player performance varies across playing conditions.

Another extension is a fully contextual, or situational, version of EDGE. In the current formulation, score context is incorporated when estimating tour-wide event weights, but the resulting fixed weight is then applied to every occurrence of the same event type in the player-level calculation. A contextual EDGE could instead retain the score-specific GE change associated with each event occurrence. For example, an ace at deuce would directly contribute more to a player EDGE than an ace at 40–0 because of its larger effect on GE. Comparing fixed-weight and contextual EDGE could help distinguish measures intended to characterize general performance from those intended to describe performance in specific high- or low-leverage situations.

EDGE could also be made more sensitive to changes in player performance over time. The current experiments aggregate MCP observations across multiple years. Future versions could calculate EDGE over rolling time windows or assign greater weight to more recent matches. Such approaches could better capture changes associated with player development, injuries, career interruptions, or changes in playing style.

An important experimental extension is out-of-sample predictive validation. Since the current fixed-weight formulation is intended to provide a measure for assessing future performance, EDGE should be calculated from an earlier set of matches and evaluated against outcomes from

a later period. Possible prediction targets include future match results, tournament advancement, and subsequent WTA rankings. EDGE could also be compared with existing measures such as DR, H+B, conventional tennis statistics, and the current WTA ranking to determine whether it provides additional predictive information.

Finally, future experiments should examine the robustness and statistical stability of EDGE. Event weights and player EDGE values could be recomputed across different time periods, minimum-match thresholds, ranking groups, and subsets of the MCP data. Bootstrap or repeated-sampling methods could quantify uncertainty in event weights and player EDGE values, particularly for players represented by relatively few matches. Such analyses would help determine how much data are necessary to obtain stable EDGE estimates and whether the conclusions remain consistent across different player populations and experimental settings.

X.6 Conclusion

EDGE considers how points are won or lost, rather than simply whether they are won or lost. It assigns tour-wide average weights to point-level event types such as aces, winners, double faults, and unforced errors according to their impact on game-win expectancy and applies these weights to a player's event counts to produce a single measure of player performance. Unlike conventional box-score statistics, which report individual point-level outcomes such as aces, winners, and unforced errors separately, EDGE combines these outcomes into a single performance measure. It also differs from existing composite metrics such as DR and H+B by incorporating score context into event valuation. Unlike the tour ranking, EDGE evaluates what happens within points and matches rather than ranking players according to tournament advancement and tournament category.

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Biography

Hanna Suzuki is a high school junior who is passionate about music, tennis, and coding. She co-founded PhysTech in 2024 and co-chaired its 2025 and 2026 editions. Hanna is a varsity tennis player for her school, and her team finished as the runner-up in the 2026 Massachusetts Interscholastic Athletic Association (MIAA) Division 2 state championship. Her recent research spans astronomical sonification, which combines coding, music, and astronomy to translate visual astronomical information into sound, and microeconomic modeling of pet ownership, which uses mathematical and computational models to explore the benefits and costs of pet ownership. Her work has earned prestigious awards, including a Grand Prize and an ACM (Association for Computing Machinery) Special Award at the 2026 International Science & Engineering Fair, the MathWorks First Place Award at the 2026 Massachusetts Science & Engineering Fair, and two best presentation awards at the International Young Researchers' Conference in 2025 and 2026. For her leadership and public service through PhysTech and other student-led technology communities, Hanna has received the Congressional Award Gold Medal (2026) and the President's Volunteer Service Gold Awards (2025, 2024 and 2023). Her technical

and research work can be found at **`https://github.com/HSSBoston`**, and her LinkedIn profile is at **`https://www.linkedin.com/in/hanna-suzuki-/`**.